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Multi-Agent System for Smart Roll-on/Roll-off Terminal Management: Orchestration and Communication Strategies for AI-Driven Optimization

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Abstract: This study presents a structured multi-agent system (MAS) architecture aimed at optimizing operational efficiency in roll-on/roll-off (Ro-Ro) terminal management through intelligent coordination and decentralized decision-making. The proposed framework enhances space allocation, route planning, traffic control, and boarding coordination, enabling real-time decision-making and adaptive operational strategies. Through structured MAS architecture, agents interact dynamically to optimize vehicle flow, reducing congestion and improving overall efficiency. The study evaluates the system's potential benefits compared to traditional port management models, highlighting improvements in transit time reduction, resource utilization, and operational resilience. The findings suggest that MAS-based automation can enhance decision-making, sustainability, and integration with Industry 4.0 paradigms, driving the transition toward intelligent, efficient, and scalable port logistics.

Keywords: multi-agent system; artificial intelligence; Ro-Ro terminals; smart ports; operational optimization; Industry 4.0

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1. Introduction

In recent decades, digitalization has become a key driver in optimizing logistics processes, transforming port management worldwide. As maritime terminal operations become more complex and the need for greater efficiency and sustainability in freight transport grows, advanced technologies within the Industry 4.0 framework have gained increasing importance [1].

Managing roll-on/roll-off (Ro-Ro) terminals, where vehicles must be efficiently organized and loaded, presents several challenges, including space allocation, internal route optimization, and the coordination of loading flows. In this context, digitalization provides innovative solutions that enhance decision-making and improve operational efficiency through process automation [2]. Maritime terminal operations, particularly in roll-on/roll-off (Ro-Ro) terminals, present complex logistical challenges where optimizing vehicle flow and space allocation is critical for operational efficiency.

One of the most promising advancements in port digitalization is the use of multi-agent systems (MAS), an artificial intelligence (AI)-based technology that enables multiple autonomous entities to interact and coordinate efficiently [3]. These systems have proven highly effective in logistics and transportation, as they allow for decentralized

decision-making, reduce reliance on human operators, and adapt to unexpected operational changes. In port environments, intelligent agents can improve vehicle distribution, reduce waiting times, and optimize internal traffic, ultimately streamlining cargo handling operations [4].

Beyond operational improvements, adopting intelligent agents in ports aligns with sustainability efforts and environmental goals [5]. Optimized traffic flow and route planning can help reduce energy consumption and lower CO₂ emissions [6]. Additionally, automating the planning and distribution of vehicles minimizes idle times and improves port infrastructure utilization, leading to reduced operational costs and greater competitiveness for Ro-Ro terminals in an increasingly demanding market [7].

Despite significant progress in digitalization, many terminals still rely on traditional management systems, which often struggle with flexibility, responsiveness, and efficiency [8]. Current space allocation and route planning models are typically based on fixed rules, making it difficult to adjust to fluctuating demand, unexpected congestion, or delays in vessel arrivals. In contrast, multi-agent systems can continuously analyze terminal conditions and adapt operations in real time, enhancing efficiency and resilience [8].

This paper aims to propose a conceptual and architectural model for multi-agent system (MAS) deployment in Ro-Ro terminal management. Given the complexity and variability of real port environments, establishing a theoretical foundation is a necessary preliminary step before undertaking pilot implementations or simulation validations.

This study aims to develop and evaluate a multi-agent model to optimize Ro-Ro terminal operations, focusing on key aspects such as space allocation, route planning, traffic management, and synchronized loading. Unlike conventional approaches, the proposed system leverages autonomous agents to improve coordination and adapt to changing port conditions. To assess its effectiveness, a comparative analysis will be conducted, measuring the system's performance against traditional management strategies in terms of reducing operational times and optimizing resource use.

This research defines a conceptual framework based on digitalization and artificial intelligence to systematically improve vehicle flow, reduce operational delays, and increase efficiency in Ro-Ro terminal logistics. The integration of intelligent agents introduces a scalable approach to logistics management, enhancing automation and responsiveness in dynamic port environments.

This study contributes to the existing literature by proposing a novel multi-agent system (MAS) architecture tailored specifically for roll-on/roll-off (Ro-Ro) terminal management. The proposed model integrates real-time route optimization, dynamic space allocation, and coordinated traffic management strategies, which have not been previously combined in this particular operational context. By addressing the complexity and dynamic conditions inherent in Ro-Ro terminals, this work offers an innovative framework aligned with the objectives of Industry 4.0 for smart port operations.

2. State of Knowledge

2.1. Industry 4.0 and Digital Transformation in Ports

Industry 4.0 has significantly transformed port logistics and operations by integrating advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), big data, and automation [9]. These innovations have made resource management more efficient, reducing operational costs and improving process traceability. In the port sector, digitalization has optimized critical operations such as space allocation, route planning, and internal traffic management, ultimately enhancing terminal efficiency [10].

One of the key benefits of digital transformation in ports is the ability to integrate smart systems that analyze real-time data [6,11]. IoT connectivity and large-scale data collection help anticipate congestion, optimize resource allocation, and improve operational

safety [12]. Additionally, digitalization enables the automation of repetitive tasks, reducing human intervention and minimizing errors in port management [10].

In Ro-Ro terminals, Industry 4.0 technologies streamline vehicle flows using intelligent algorithms. Advanced synchronization of operations helps reduce waiting times and maximizes port infrastructure efficiency [13]. However, challenges remain, including interoperability between different systems, cybersecurity risks, and resistance to change in an industry that has traditionally relied on manual or semi-automated processes.

2.2. AI Applications in Port Management

Artificial intelligence (AI) is increasingly being adopted to enhance port operations, particularly in planning and optimization [14]. Machine learning algorithms can predict traffic demand, improve infrastructure utilization, and minimize operational delays. AI has proven particularly useful in identifying congestion patterns and enhancing security through automated anomaly detection [15].

In traffic management and space allocation, AI-based systems analyze vast amounts of data in real time, enabling faster and more accurate decision-making [16]. For example, optimization algorithms can automatically assign parking areas for waiting vehicles, improving Ro-Ro terminal efficiency. Similarly, neural networks and predictive models help anticipate operational disruptions and adjust logistics dynamically.

Despite its advantages, implementing AI in ports faces some challenges. The reliability of AI models depends on data quality and availability, and integrating AI with existing port management systems can be technically complex [17]. Moreover, automated decision-making requires a balance between system autonomy and human oversight, particularly in critical situations that demand immediate intervention [18].

2.3. Optimization of Vehicle Movements in Ro-Ro Terminals

Ro-Ro terminals face unique logistical challenges due to the constant movement of vehicles and the need for optimized internal flows [19]. Efficient management of vehicle movements within the terminal is crucial to avoiding congestion, minimizing waiting times, and streamlining loading operations. To address this, advanced route planning models and smart technologies have been introduced to improve space allocation and traffic flow [13].

Several approaches have been proposed to enhance mobility in Ro-Ro terminals, including search algorithms such as A*, Dijkstra, and reinforcement learning techniques [20]. These models calculate optimal routes in real time by considering factors such as available space, distance to the boarding point, and vehicle departure priority. Additionally, IoT integration allows for dynamic route adjustments based on real-time terminal traffic conditions [21].

Despite advancements in movement optimization, many terminals still rely on static planning models, limiting their adaptability to operational changes. A lack of coordination among different port logistics agents often leads to inefficiencies, increasing waiting times and reducing overall capacity [22]. Implementing a multi-agent system to optimize coordinated vehicle movements could significantly improve efficiency in these terminals.

2.4. Research Gap

While port digitalization and AI implementation have advanced, there are still limitations in applying multi-agent models to optimize vehicle movements in Ro-Ro terminals [13]. Most studies focus on optimizing specific processes, such as space allocation or route planning, but few have integrated these elements into a coordinated autonomous system [23,24].

Additionally, many terminals continue to use traditional methodologies or optimization systems that lack real-time adaptability [25]. Current models struggle to respond efficiently to dynamic port conditions, such as fluctuating boarding demand or unexpected congestion. Furthermore, inadequate communication between different port systems hinders overall vehicle flow optimization.

This study aims to bridge these gaps by developing a multi-agent system for Ro-Ro terminal management, enabling decentralized decision-making, optimized space and route allocation, and improved coordination among operational agents. By integrating these components into a single autonomous system, the research seeks to enhance port efficiency, aligning Ro-Ro terminal management with Industry 4.0 principles.

3. Methodology Steps

Section 3 details the structured methodological approach followed to design and propose a multi-agent system (MAS) tailored to the operational characteristics of roll-on/roll-off (Ro-Ro) terminal management. While grounded in MAS design principles, the methodology is specifically adapted to address the dynamic operational needs of maritime terminals.

The methodology steps were defined based on standard practices in the design of intelligent multi-agent systems for logistics optimization, adapted to the specific operational conditions of Ro-Ro terminals. The methodology leverages heuristic algorithms, IoT infrastructure, and decentralized coordination protocols. Potential challenges include data integration, system interoperability, resistance to change among operators, and computational resource requirements.

To develop an optimized management system for Ro-Ro terminals, a structured methodological approach was followed. The methodology is based on a multi-agent system (MAS) framework, integrating digitalization and artificial intelligence techniques to enhance operational efficiency (Figure 1). The process consists of six key steps, from defining the system's scope and requirements to evaluating its theoretical benefits compared to traditional port management models. Each phase ensures a structured design, selection of appropriate algorithms, modeling of agent communication, and integration of human interaction to achieve a scalable and adaptable solution.

While the methods employed draw from established multi-agent system (MAS) design principles, the specific integration and orchestration of intelligent agents to manage the operational flow of Ro-Ro terminals, as structured in Figure 1, represent a novel adaptation that has not been formally documented in previous port management research.

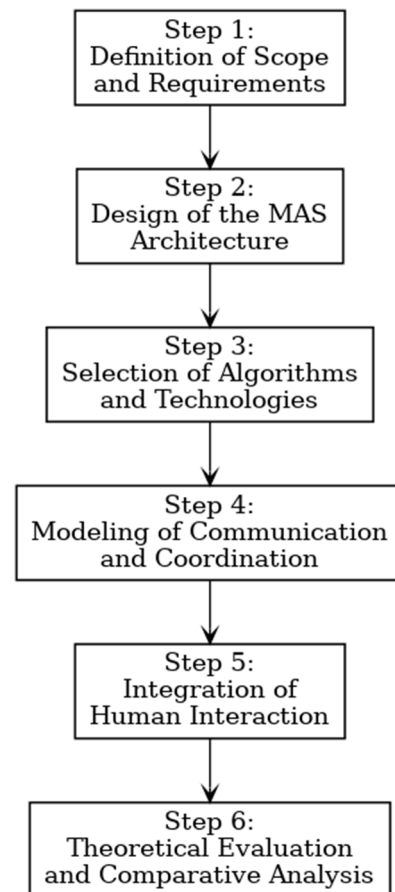


Figure 1. Methodology steps.

3.1. Defining the Scope and System Requirements

The objectives and needs of the multi-agent system for the optimization of management in Ro-Ro terminals are established. In this phase:

1. Key port processes that can benefit from digitalization and intelligent automation are identified.
2. The limitations of traditional vehicle management systems in Ro-Ro terminals are analyzed.
3. The functional and non-functional requirements of the system are defined, including efficiency, reduction of transit times, energy optimization, and scalability.

3.2. Multi-Agent System (MAS) Architecture Design

A structured design of the multi-agent system architecture multi-agent system (MAS) model is developed, including:

1. Definition of intelligent entities and their respective functions (orchestration, space allocation, route optimization, flow control, boarding management, and supervision).
2. Specification of communication between agents, defining information exchange protocols, and collaborative decision-making models.
3. Development of an architecture diagram, visualizing the key interactions between agents to ensure an efficient flow of information and operations.

The integration logic between heterogeneous optimization algorithms is managed by the orchestrator agent (EOO), which coordinates task allocation and data exchange among specialized agents through standardized communication protocols such as contract net protocol (CNP). Each agent operates its assigned optimization function autonomously,

minimizing the need for centralized processing. Regarding computational resources, the system design is distributed to enable scalability, with each agent requiring moderate processing capabilities suitable for typical industrial IoT or edge computing environments.

3.3. Selection of Optimization Algorithms and Technologies

The most appropriate techniques and tools are identified and selected for each entity in the system:

1. Heuristic and metaheuristic optimization for space allocation (genetic algorithms, tabu search).
2. Route planning algorithms (A*, Dijkstra, reinforcement learning) to minimize transit times.
3. Traffic analysis and simulation for flow control using IoT sensors.
4. Scheduling models and dynamic planning for the coordination of vehicle boarding.
5. Artificial intelligence and predictive analytics for system monitoring and continuous improvement.

3.4. Modeling of Communication and Coordination Between Agents

A structured communication model is designed, establishing efficient data exchange strategies:

1. Hierarchical communication with the orchestrating agent (EOO) for global decisions.
2. Direct interaction between specialized agents to minimize latencies.
3. Negotiation based on priorities and availability in the allocation of resources.
4. Use of middleware and standard communication protocols (JADE, MQTT, ROS) to ensure interoperability in digital environments.

3.5. Integration of Human Interaction into the System

The level of human intervention within the MAS is defined, ensuring that port operators can interact with intelligent entities when necessary:

- Supervisors can modify priorities and reassign spaces in the terminal.
- Controllers can intervene in route planning and traffic management in critical cases.
- Managers can evaluate performance metrics and make strategic adjustments to the system.

3.6. Theoretical Evaluation and Comparative Analysis

A theoretical evaluation is carried out based on:

- Comparison with traditional port management models, measuring potential benefits of the system.
- Analysis of the expected impact on the reduction of time and energy consumption in Ro-Ro terminals.
- Identification of challenges and limitations in the implementation of the multi-agent system.

4. Results and Discussion

The results presented in this study directly apply the methodological framework described in the previous section. Each finding is connected to a specific methodological step, ensuring that the improvements proposed in space allocation, route optimization, traffic control, and boarding management are systematically supported by the multi-agent system architecture developed.

The evaluation of the results obtained in this study is structured according to the methodological steps previously defined. Each finding responds to a specific phase of the

development of the multi-agent system, from the identification of needs to the benchmarking with traditional models. The main results are presented below, pointing out the relationship between the methodology applied and the benefits obtained in terms of operational efficiency, route optimization, reduction of transit times, and improvements in the synchronization of shipment.

The results obtained from the design and analysis of the multi-agent system for the optimization of management in Ro-Ro terminals reflect a series of theoretical and operational benefits compared to traditional management models. The main findings are presented below and organized according to the defined methodology.

A Ro-Ro terminal is designed to manage vehicles and Ro-Ro units that are loaded and disembarked using their own wheels, thus optimizing cargo handling and reducing operating times compared to other loading systems. The main activities and processes that characterize the operation of these terminals are described below and structured in stages.

The operation of a Ro-Ro (Roll-on/Roll-off) terminal follows a structured flow that allows the efficient loading and unloading of vehicles. This process is made up of three main stages:

Arrival and parking in waiting areas: vehicles enter the terminal and are directed to designated areas where they await their turn for loading or unloading. At this stage, documentation is checked, and traffic is organized.

Loading/unloading: vehicles are guided to or from the ship using their own wheels. This occurs through specially designed ramps that connect the dock to the vessel.

Storage or departure: After disembarking, vehicles are directed to temporary storage areas or directly outside the terminal, depending on their final destination.

The use of technologies such as the Internet of Vehicles (IoV) and automation improves coordination, reduces waiting times, and optimizes vehicular flow at each stage (Figure 2). This diagram provides contextual understanding for readers unfamiliar with the sequential logistics of vehicle movement within the terminal and is not part of the proposed MAS model.

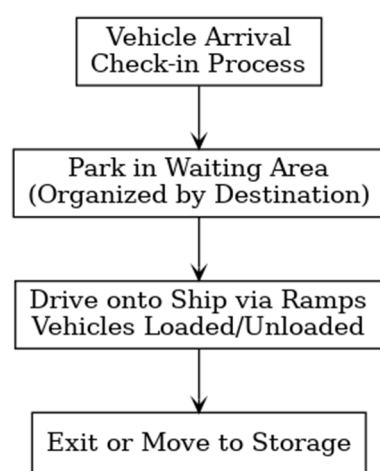


Figure 2. Simplified schematic of the basic operational flow in a roll-on/roll-off (Ro-Ro) terminal. Source: Own elaboration.

4.1. Definition of the Scope and System Requirements

From the analysis of port processes in Ro-Ro terminals, the following problems were identified in traditional systems, both in international literature and during on-site visits to study terminals of the Spanish port system. These inefficiencies are especially relevant in the current context of port digitalization, which is guided by the foundational principles of Industry 4.0—namely, digital integration, real-time decision-making, interoperability,

data-driven process optimization, and autonomous system coordination. However, most traditional Ro-Ro terminals still operate under fragmented, reactive systems that fall short of these principles [1,2].

The identification of operational inefficiencies was based both on a review of international literature and on observations gathered during technical visits to several Ro-Ro terminals within the Spanish port system. These visits provided first-hand insights into the current limitations of traditional management approaches:

- Lack of automation in the allocation of spaces, which generates inefficiencies and unnecessary waiting times.
- Non-optimized routes within the terminal, causing longer journeys and higher energy consumption.
- Reactive traffic management, without real-time adjustment mechanisms.
- Manual coordination of shipment, which introduces delays and reduces operational capacity.
- Low data integration and centralized supervision, making strategic decision-making difficult.
- Manual space assignment, without location optimization.
- Uncoordinated flow of vehicles, with inefficient routes.
- Internal congestion in the field due to lack of dynamic regulation.
- Manual boarding processes, generating unnecessary waiting.
- Reactive monitoring, based on late reports instead of real-time data.

The proposed multi-agent system addresses these challenges by integrating distributed artificial intelligence, enabling decentralized and optimized management of vehicles within the terminal.

4.2. Design of the Multi-Agent System Architecture (MAS)

The system is made up of six intelligent entities that manage different aspects of port operations:

4.2.1. Intelligent Entities: Function, Required Data, and Human Interaction

The design of the multi-agent system (MAS) is based on the integration of various intelligent entities, each of them responsible for managing a specific aspect of the port operation (Table 1). These entities work in a coordinated manner to optimize efficiency and decision-making within the port environment, ensuring more dynamic and automated management. The main intelligent entities of the system are presented below, along with their respective functions, the data they require for their operation, and the level of interaction they maintain with users.

Table 1. Intelligent entities of the multi-agent system: function, required data, and human interaction.

Entity	Main Function	Required Data	Human Interaction
EOO—Operations Orchestrating Entity	Coordinates communication and synchronization between entities. Make strategic decisions and resolve conflicts in real time.	Terminal status (occupied spaces, traffic, ramp availability). Operational reports of each entity.	Supervisors can manually modify priorities and approve critical decisions. Operators can report incidents.
EDV—Vehicle Distribution Entity	Determines the best location for each vehicle in the field, reducing travel times.	Availability of spaces in the field. Type and dimensions of the vehicle. Estimated boarding time.	Operators can manually reassign locations in special cases.

ENR—Navigation and Routes Entity	Calculate optimal routes to minimize transit times and energy consumption.	Current location and destination of each vehicle. Internal traffic status.	Operators can modify routes in the event of blockages or adverse conditions.
ECF—Flow Control Entity	It regulates traffic within the terminal to avoid congestion and high waiting times.	Traffic status at the terminal. Movement restrictions.	Controllers can manually activate restrictions or release locked zones.
ECE—Shipping Coordination Entity	Manage the arrival of vehicles at the boarding ramp, synchronizing the flow with the departure of the ship.	Departure time of the vessel. Capacity and availability of the boarding ramp. Condition of vehicles in transit.	Supervisors can modify boarding shifts. Managers can adjust load strategies.
ESO—Supervisory and Optimization Entity	Evaluate system performance and adjust operating parameters to improve efficiency.	Operation KPIs of each entity. Performance history and data analysis.	Managers can review efficiency reports and make strategic adjustments.

4.2.2. Selection of Algorithms and Technologies for Optimization

The selection of the algorithms used in the model is based on their ability to address the specific challenges of port management in Ro-Ro terminals. Table 2 presents the advantages and disadvantages of each of these algorithms in the port context, highlighting their applications within the proposed multi-agent system. The integration of these algorithmic approaches supports adaptive route planning, dynamic space allocation, and decentralized traffic coordination aligned with terminal conditions.

Table 2 provides a qualitative comparison of optimization approaches relevant to intelligent port management. It does not constitute direct algorithmic benchmarking.

Table 2. Comparison of algorithms in the port context.

Algorithm	Advantages	Disadvantages	Application in the Model
A* (Route Search)	Find the optimal route with computational efficiency. Adaptable to real-time traffic changes.	It can become computationally expensive in environments with many variables.	Optimization of routes within the terminal.
Dijkstra (Route Search)	Exact algorithm to find the shortest route. Robust and widely validated.	Less efficient compared to A* in environments with large volumes of data.	Alternative for trajectory planning.
Reinforcement Learning Techniques	Able to learn dynamically and improve decisions with experience.	It requires prior training and large volumes of data to improve its accuracy.	Optimization of internal traffic management.
Contract Net Protocol (CNP)	It provides a decentralized mechanism for trading between agents.	It can generate more communication traffic between agents, affecting latency in large systems.	Coordination of tasks between agents (spaces, routes, traffic).
Taboo Search (Allocation of Spaces)	Find optimal solutions to complex combinatorial problems.	It does not guarantee finding the best global solution, but it avoids falling into local lows.	Optimization of the allocation of spaces in the field.
Genetic Algorithms (Allocation Of Spaces)	Ability to optimize in large search spaces. Find solutions close to the optimum.	It requires longer processing time and parameter adjustment for optimal performance.	Alternative for the allocation of vehicle spaces in the terminal.

The algorithms used for each intelligent entity allow port processes to be optimized (Table 3):

Table 3. Justification of the main algorithms by intelligent entity.

Entity	Main Algorithm	Justification
EOO—Operations Orchestrating Entity	Multi-agent systems with contract net protocol (CNP)-based trading	An efficient coordination mechanism is required between entities with different objectives and constraints. The contract net protocol (CNP) allows specialized agents to negotiate tasks without overloading a single central node. In addition, it guarantees flexibility in the face of changes in demand and distribution of resources.
EDV—Vehicle Distribution Entity	Heuristic optimization (taboo search, genetic algorithms)	The optimal allocation of spaces in the field is a problem of combinatorial optimization. Advanced heuristics, such as taboo search and genetic algorithms, make it possible to find solutions close to the optimal without the need to evaluate all possible combinations, significantly reducing calculation time.
ENR—Navigation and Routes Entity	Pathfinder A, Dijkstra, reinforcement learning	To find the shortest path from the field to the boarding ramp, route planning algorithms are required. Dijkstra ensures the optimal path in networks with positive weights, while A* improves efficiency in dynamic environments with heuristics. If traffic is highly changeable, you can use reinforcement learning, which adapts routing decisions based on previous experiences.
ECF—Flow Control Entity	Traffic models with IoT and machine learning	Real-time traffic management involves handling sensor data and predicting congestion patterns. Machine Learning models (regression, neural networks) can anticipate areas of high congestion and suggest proactive adjustments in traffic flows, avoiding bottlenecks.
ECE—Shipping Coordination Entity	Dynamic scheduling with metaheuristics	Vehicle coordination on the boarding ramp must be adapted based on the load, vessel restrictions, and ramp availability. Algorithms such as GRASP (Greedy Randomized Adaptive Search Procedure) and Colony Optimization optimize shipment time planning with computational efficiency.
ESO—Supervisory and Optimization Entity	Machine learning for anomaly detection and optimization	To continuously evaluate the efficiency of the system, it is necessary to analyze historical patterns and detect anomalies in operations. Unsupervised learning models (clustering, outlier detection with DBSCAN or Isolation Forest) allow inefficiencies to be identified without manual intervention.

The selection of these algorithms responds to the need to balance accuracy, computational efficiency, and adaptability in real time. Each entity requires a specific approach:

- Heuristic optimization in combinatorial problems.
- Efficient route search in dynamic environments.
- Data-driven predictive modeling in traffic management and monitoring.
- Multi-agent negotiation for decentralized coordination.

The rationale for the Core Algorithms is detailed below:

- Efficient and flexible coordination: the use of contract net protocol (CNP)-based trading in the EOO allows agents to manage tasks in a decentralized manner, reducing overhead on a single node and ensuring adaptability to changes in demand.
- Rapid optimization in space allocation: the taboo search and genetic algorithms in the EDV allow efficient solutions to be found without the need to evaluate all possible combinations, achieving a better distribution of vehicles in the field with reduced processing times.
- Dynamic and efficient routes: the use of A*, Dijkstra, and reinforcement learning in the ENR allows optimal routes to be calculated by adapting to changes in traffic, ensuring shorter and more efficient journeys within the terminal.

- Intelligent traffic management: machine learning with predictive models in the ECF allows congestion to be anticipated and traffic to be dynamically adjusted in the terminal, avoiding blockages and downtime in operations.
- Accurate boarding synchronization: the application of metaheuristics for dynamic scheduling in the ECE improves the allocation of boarding shifts, reducing delays and optimizing ramp utilization.
- Monitoring and continuous improvement: the use of machine learning to detect anomalies in ESO allows for identifying patterns of inefficiency and optimizing the system based on historical data, ensuring progressive improvements in port management.

These algorithms ensure that the multi-agent system (MAS) operates with efficiency and scalability, enabling autonomous and optimized decision-making at the Ro-Ro terminal.

4.2.3. Modeling Communication and Coordination Between Agents

To ensure efficiency, a structured communication model was designed with three levels:

1. Hierarchical (EOO coordinates all entities).
2. Direct interaction between specialized entities.
3. Negotiation based on priorities for the distribution of resources.

Efficient communication between entities ensures coordinated and adaptable management to operational changes in the terminal (Table 4).

Table 4. Communication between entities.

Entity	Send Data To	Receive Data From
EOO (Orchestrator)	ALL	ALL
EDV (Vehicle Distribution)	EOO, ENR	EOO
ENR (Navigation and Routing)	EOO, ECF	EDV, EOO
ECF (Flow Control)	EOO, ECE	ENR, EOO
ECE (Shipping Coordination)	EOO, ESO	ECF, EOO
ESO (Monitoring and Optimization)	EOO	ALL

The orchestrator coordinates and exchanges information with all specialized agents, which also interact with each other based on operational dependencies. The hierarchical layout reflects the functional relationships and data flows necessary for coordinated terminal operations (Figure 3).

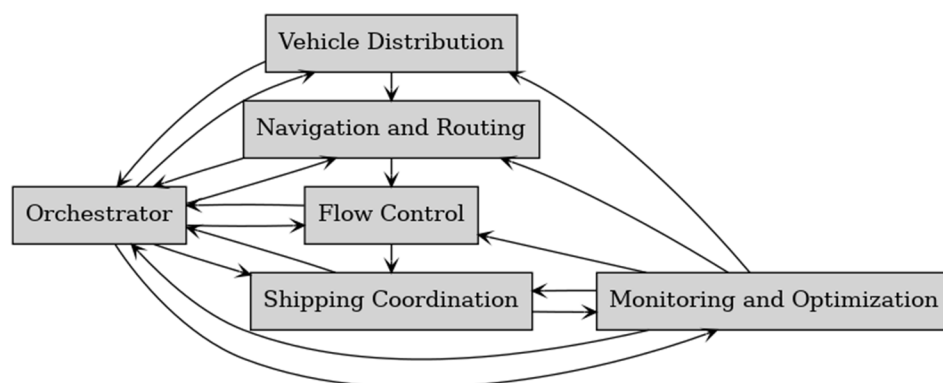


Figure 3. Communication flow between intelligent agents within the proposed MAS architecture. Source: Own elaboration.

The architecture of the model would be:

1. **Hierarchical Architecture (Hierarchical):** the system has an orchestrating agent (EOO) that manages and coordinates the other agents, which resembles the hierarchical structure of the image. Decision-making flows from the top down, with each agent specializing in specific tasks.
2. **Router + Aggregator Pattern:** The EOO acts as a router that receives data from different agents and redistributes it according to operational needs. It also works as an aggregator, collecting information from multiple sources (IoT sensors, optimization algorithms, traffic data) to generate coordinated decisions.
3. **Shared Database with Different Tools:** The model is based on the integration of optimization tools, planning algorithms, and operational databases, which fits with the pattern of a shared database used by different agents.

The multi-agent system is a combination of a hierarchical architecture with elements of router, aggregator, and shared database. The orchestrator agent centralizes decision-making, but specialized agents interact with different tools to optimize traffic, space allocation, and boarding.

4.3. Integration of Human Interaction into the System

Although the multi-agent system (MAS) automates much of the port management, it is essential to allow human intervention at different levels to ensure operational flexibility and responsiveness to unforeseen events.

Three levels of interaction were established within the system (Table 5):

1. **Supervisors**
They adjust orchestrator agent (EOO) parameters to modify priorities.
They review alerts generated by the monitoring and optimization agent (ESO).
They can approve or block critical system decisions.
2. **Operators in the terminal**
They can modify the allocation of spaces in the field managed by the vehicle distribution agent (EDV).
They intervene in the modification of routes generated by the navigation and routing agent (ENR) if there are unexpected blockages.
They make adjustments to flow control agent (ECF) traffic restrictions in cases of severe congestion.
3. **Terminal managers**
Accesvaluate performance metrics and operational efficiency.

Table 5. Human interaction by agent.

Entity	Permitted Human Intervention	Level of Intervention
EOO (Orchestrator)	Modification of priorities and operational rules.	Supervisors
EDV (Vehicle Distribution)	Manual reassignment of spaces in the field.	Terminal operators
ENR (Navigation and Routing)	Route adjustment and unlocking of closed areas.	Terminal operators
ECF (Flow Control)	Application of restrictions or release of roads.	Terminal operators
ECE (Shipping Coordination)	Rescheduling of boarding times.	Supervisors and managers
ESO (Monitoring and Optimization)	Analysis and redefinition of strategies.	Terminal managers

Among the benefits of human integration are greater flexibility in strategic decision-making, better adaptability to unforeseen events such as ship delays or congestion, and the assurance of operational supervision without relying 100% on automation.

This integration ensures that the MAS does not replace humans but rather enhances the responsiveness and optimization of the terminal.

Finally, the multi-agent model could be represented as shown in Figure 4.

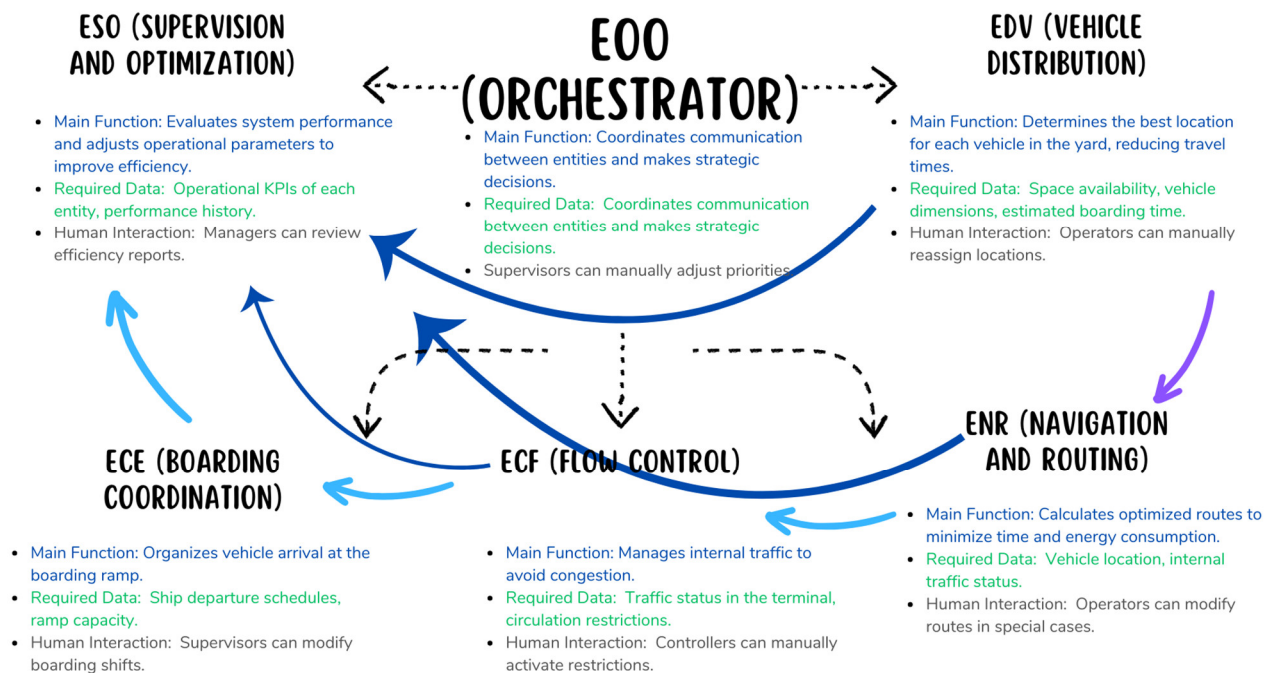


Figure 4. A complete multi-agent system.

Figure 4 illustrates the communication flow and functional dependencies among the intelligent agents defined in the proposed MAS architecture. The orchestrator agent acts as the central coordinator, receiving and distributing information to all other agents. Vehicle distribution, navigation, and routing agents interact closely to define optimal parking locations and internal routes. Flow control manages congestion based on routing outputs and relays restrictions to shipping coordination, which schedules boarding accordingly. Monitoring and optimization collects performance indicators from all agents and provides system-level feedback to both the orchestrator and other agents, enabling adaptive behavior. The diagram highlights not only bilateral communication but also vertical dependencies, where decisions by downstream agents depend on the orchestration and data consolidation performed at higher levels. This structure enables scalable coordination across operational layers of a Ro-Ro terminal.

4.4. Theoretical and Comparative Evaluation of the Model

Various studies have shown that the implementation of systems based on artificial intelligence and heuristic optimization in logistics environments can generate significant improvements in operational efficiency [6,11,26]. Previous research has shown that the use of multi-agent models in port infrastructure management can reduce waiting times and improve the use of available resources [26]. For example, in automated port terminals, the use of route planning algorithms such as A* and Dijkstra has made it possible to reduce transit times by 15–30% compared to traditional models based on fixed rules [27]. Likewise, the integration of dynamic scheduling systems in Ro-Ro terminals has shown that the optimization of boarding times can reduce downtime in the loading and unloading flow by up to 20% [28]. These findings support the feasibility of the approach proposed

in this study, providing a solid theoretical basis for comparison with traditional port management models.

Below in the Table 6 shows a comparison with the expected theoretical metrics.

Table 6. Comparison multi-agent model vs. traditional model.

Criterion	Traditional Model	Multi-Agent Model (Proposed)	Reference
Allocation of Spaces	Manual or semi-automatic, with high occupancy times.	Dynamic optimization with heuristics, expected 25% reduction in occupancy times.	[29]
Route Optimization	Static, based on operational experience.	Dynamic calculation with A*, Dijkstra, and reinforcement learning, expected reduction of 15–30% in transit times.	[30]
Traffic Management	Reactive, without real-time adjustments.	Continuous adaptation based on prediction and IoT sensors.	[31]
Synchronized Boarding	Based on fixed schedules, without dynamic optimization.	Dynamic adjustment with adaptive scheduling, 20% reduction in waiting times.	[32]
System Monitoring	Based on manual reports and subsequent adjustments.	Predictive analytics and real-time optimization using AI.	[33]

A theoretical comparison was made with traditional port management models, showing improvements in:

- Space allocation: Reduction of occupancy time by up to 25%.
- Route optimization: Less unnecessary travel and energy savings.
- Traffic management: Dynamic adaptation to congestion.
- Synchronized boarding: Avoid unnecessary waiting.
- Proactive monitoring: Reduction of operational errors through AI.

The theoretical analysis suggests that the proposed multi-agent system has the potential to improve the operational efficiency of the Ro-Ro terminal, aligning with the principles of Industry 4.0.

These principles include digitalization, real-time decision-making, interoperability, and autonomous management [1].

The evaluation of the multi-agent model has been carried out through a theoretical comparative analysis based on three key approaches: (1) the comparison with previous studies on optimization in logistics and port terminals, (2) the evaluation of standard times and processes in traditional port management models, and (3) the validation of the algorithms used, whose efficiency has been demonstrated in various logistics applications. First, the literature supports that the use of heuristic algorithms in the allocation of spaces can significantly reduce the occupancy time in port fields. Likewise, route optimization using algorithms such as A* and reinforcement learning has shown improvements in energy efficiency and reduced travel times in logistics environments. In addition, the use of predictive models for traffic management and boarding synchronization allows dynamic adaptation to congestion and operational schedules, reducing unnecessary waiting and operational errors. These improvements, based on previously validated optimization methodologies, provide a solid basis for theoretical comparison and justify the applicability of the multi-agent system in improving the operational efficiency of Ro-Ro terminals within the framework of Industry 4.0.

The proposed multi-agent system presents a viable approach to the intelligent management of Ro-Ro terminals; however, its implementation in a real environment requires the consideration of various operational and technological factors. Firstly, the adoption of this model requires integration with existing port management systems, which implies compatibility with digital infrastructures and operational planning platforms. Many

terminals operate with legacy systems that might require upgrades or interoperability modules to enable seamless communication between smart agents and port traffic control systems.

Another key challenge is the technological infrastructure needed for real-time data collection and processing. Optimal system operation requires the implementation of IoT sensors, advanced communication networks, and servers capable of handling distributed information processing. Although many terminals have already adopted digital technologies, heterogeneity in automation levels between different ports can represent a barrier to model standardization.

From an operational point of view, the interaction between the autonomous system and port operators is essential to ensure the adaptability and efficiency of the model. Staff training and redefining roles within the terminal can facilitate the transition to AI-based management. In addition, the system's acceptance will depend on its ability to demonstrate tangible improvements in reducing operating times, optimizing internal traffic, and efficiency in shipment planning. In this sense, validation through pilot projects in controlled environments would allow the performance of the model to be evaluated before its large-scale deployment.

4.5. Limitations and Future Lines of Work

This study proposes a conceptual architecture for the application of a multi-agent system (MAS) in roll-on/roll-off (Ro-Ro) terminal management. The model focuses on defining the agents' roles, coordination protocols, and expected benefits in terms of operational optimization. However, the work does not include experimental results, simulation-based testing, or empirical data.

The lack of validation is a recognized limitation. The implementation of a MAS in real port environments requires significant institutional coordination, access to data, and technical resources, which are often restricted by budgetary and regulatory constraints. While several port authorities are actively exploring digitalization and AI-based solutions, large-scale pilots involving intelligent agents remain limited.

Future research will focus on building simulation environments that replicate terminal operations and agent interactions under controlled conditions. In parallel, collaboration with operational terminals will be sought to deploy small-scale prototypes or pilot studies. These efforts will aim to evaluate the model's performance using operational indicators such as efficiency gains, reduced vehicle dwell time, and adaptability to real-time conditions.

5. Conclusions

The implementation of a multi-agent system in the management of Ro-Ro terminals represents a significant advance in the digitalization and optimization of port operations. Throughout this study, a model has been developed that allows for the dynamic allocation of spaces, the optimization of internal routes, and the proactive management of traffic and boarding. Unlike traditional approaches, the proposed system integrates artificial intelligence and advanced heuristic algorithms to improve operational efficiency and reduce transit times within the terminal.

The comparative analysis carried out in this study has shown that the digitalization of these processes can generate tangible improvements in port operations. It has been identified that the implementation of intelligent agents can reduce the time it takes to occupy spaces in the field by up to 25%, minimize unnecessary trips through the use of dynamic routes, and improve the ability of internal traffic to adapt to variations in operational demand. In addition, intelligent boarding synchronization reduces waiting times

and optimizes ramp capacity, contributing to reduced costs and increased terminal productivity.

From a technological perspective, the results obtained confirm the feasibility of using multi-agent systems in port environments. The integration of algorithms such as A*, Dijkstra, and reinforcement learning for route optimization, heuristic models for space allocation, and multi-agent negotiation strategies such as contract net protocol (CNP), have proven to be effective tools in decentralized decision-making. These advances position artificial intelligence as a key factor in the evolution toward more efficient and sustainable ports.

Despite the benefits identified, the implementation of these systems in real environments presents challenges. Interoperability with existing port management systems, the need for appropriate technological infrastructure, and resistance to change on the part of port operators are factors that need to be considered. In addition, human supervision remains a critical element in ensuring safety and strategic decision-making in critical operational situations.

As a future line of research, it is recommended that experimental tests and simulations be carried out to more accurately quantify the benefits of the proposed system. Validation in real environments would allow the impact of the model to be evaluated in different operational scenarios and to define scalable implementation strategies. Likewise, the use of hybrid models that combine AI with historical data analysis could further improve the predictive capacity of the system.

In conclusion, this study demonstrates that the use of artificial intelligence and multi-agent systems in Ro-Ro terminals is a viable and promising alternative to optimize port management. The application of these technologies not only improves operational efficiency but also contributes to the transition to more automated, resilient ports aligned with Industry 4.0 principles.

Although the model has not yet been validated through real-world implementation, it offers a solid conceptual basis for future experimental research, simulation studies, and pilot projects within operational Ro-Ro terminals.

While this paper does not include performance data from simulations or real deployments, the proposed model offers a structured foundation for future experimentation. The next stage of research will involve testing the system in controlled environments, either through discrete-event simulation or collaborative pilot implementations in operational terminals.

It is important to note that the system remains a conceptual model and has not yet been validated through simulation or real-world implementation. Its contribution lies in providing a structured foundation for future experimentation and applied development.

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